

LOWELL, ID, USA

WMO: 725784

Lat: 46.144N

Lon: 115.596W

Elev: 1480

StdP: 13.93

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 04109

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.2	18.5	3.2	6.8	17.8	9.8	9.6	24.6	8.9	24.5	7.0	27.3	2.6	0	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.0	94.9	67.7	91.2	66.3	87.8	65.1	70.0	88.1	68.2	86.3	66.5	83.9	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	94.8	75.2	62.6	89.6	73.4	60.7	83.6	71.6	34.8	88.2	33.2	86.7	31.9	83.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	6.6	5.6	7.8	100.2	5.9	2.9	3.6	102.3	0.1	104.0	-3.2	105.7	-7.5	107.8		
(5)			7.6	73.9	5.7	1.5	3.5	74.9	0.2	75.8	-3.1	76.6	-7.2	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	33.3	36.4	43.2	49.5	57.2	64.0	72.7	71.3	62.2	49.7	39.3	32.6		
	DBStd	14.97	5.91	6.20	5.18	5.68	6.29	6.68	5.05	4.86	5.97	5.86	5.88	5.64		
	HDD50	2145	517	382	218	77	11	1	0	0	2	75	322	540		
	HDD65	5658	982	801	675	466	250	95	8	10	121	474	771	1005		
	CDD50	2519	0	0	7	61	235	421	703	659	367	65	1	0		
	CDD65	559	0	0	0	0	9	64	245	205	36	0	0	0		
	CDH74	6549	0	0	0	32	278	883	2645	2192	512	7	0	0		
	CDH80	3047	0	0	0	3	72	339	1382	1099	152	0	0	0		
(14) Wind	WSAvg	1.6	1.6	1.5	1.6	1.7	1.6	1.7	1.9	1.7	1.5	1.2	1.5	1.5		
(15) Precipitation	PrecAvg	32.6	3.5	2.4	3.0	3.2	3.3	3.1	1.2	1.4	2.0	2.8	3.9	3.1		
	PrecMax	49.4	8.3	5.8	8.1	5.9	6.8	7.6	3.9	4.4	5.1	7.7	11.3	7.5		
	PrecMin	19.0	0.5	0.1	0.5	0.5	1.0	0.7	0.0	0.0	0.1	0.0	0.7	0.2		
	PrecStd	6.8	1.9	1.3	1.6	1.3	1.1	1.6	1.1	1.3	1.4	1.9	2.1	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	56.0	67.2	78.5	86.7	94.3	100.0	98.4	89.8	74.1	55.6	45.4		
		MCWB	42.0	46.4	51.6	57.1	61.4	68.3	68.9	67.6	66.1	59.4	49.4	42.5		
	2%	DB	43.1	50.2	61.2	72.2	81.4	88.1	96.4	94.0	84.2	66.5	51.6	42.7		
		MCWB	40.4	43.6	48.9	55.2	59.9	65.4	68.0	67.1	63.2	56.0	47.4	40.5		
	5%	DB	40.3	46.0	55.8	66.0	76.4	83.7	92.5	90.9	80.5	61.9	48.6	40.1		
		MCWB	38.0	41.2	46.6	52.1	58.3	63.1	66.9	65.9	62.0	53.1	45.3	38.4		
	10%	DB	38.4	43.0	51.7	60.7	71.2	79.0	88.6	87.1	75.3	57.3	46.4	38.1		
		MCWB	36.7	40.1	45.0	49.5	56.4	61.1	65.5	64.7	60.0	51.1	42.9	36.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	47.9	53.5	59.4	64.3	70.3	73.3	71.5	68.4	61.1	51.4	43.7		
		MCDB	45.4	53.7	63.8	74.3	80.0	87.6	90.5	90.8	83.3	69.8	53.7	45.0		
	2%	WB	40.4	44.4	50.1	55.9	61.6	67.5	70.6	69.3	65.6	57.4	48.5	41.0		
		MCDB	42.8	48.5	58.5	69.1	77.5	84.1	88.5	88.4	81.0	64.4	50.6	42.4		
	5%	WB	38.4	42.2	47.7	53.3	59.6	65.0	68.7	67.6	63.3	54.6	46.3	38.8		
		MCDB	40.1	45.4	54.6	63.6	72.7	79.8	86.8	86.4	76.7	59.8	48.2	40.0		
	10%	WB	37.0	40.2	45.7	50.8	57.6	62.7	67.1	65.8	61.0	52.5	44.2	37.1		
		MCDB	38.2	42.8	50.8	58.4	68.6	75.5	85.0	83.6	72.8	56.4	46.0	38.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	12.9	18.2	21.9	25.3	27.1	34.0	34.0	28.2	17.9	10.4	8.1		
		MCDBR	11.0	17.9	26.5	33.5	35.7	35.7	39.2	38.8	34.4	24.8	13.7	9.6		
		MCWBR	8.1	11.8	15.4	17.0	15.7	14.8	14.7	15.2	16.0	14.4	9.8	7.2		
	5% WB	MCDBR	10.1	16.5	23.6	30.2	31.4	33.7	35.4	35.9	32.5	21.9	12.5	8.6		
		MCWBR	7.6	11.2	14.6	16.2	14.5	14.8	14.4	15.3	15.8	13.1	9.1	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.287	0.328	0.382	0.407	0.401	0.411	0.455	0.408	0.339	0.339	0.279		
	taud		2.467	2.463	2.421	2.288	2.271	2.331	2.316	2.155	2.261	2.485	2.343	2.505		
	Ebn at Noon		264	285	286	277	273	275	270	252	254	261	233	253		
	Edh at Noon		21	26	31	39	41	39	39	44	36	25	24	18		
(44) All-Sky Solar Radiation	RadAvg		420	673	1030	1474	1780	2040	2346	1974	1431	853	459	322		
	RadStd		50	79	104	124	142	176	129	109	136	84	45	37		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+68

Nomenclature: See separate page